1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
D3_96509_Aditya> delimiter $$
D3_96509_Aditya> create procedure sp_area(p1 DOUBLE,p2 DOUBLE)
    -> begin
    -> declare v1 DOUBLE;
    -> declare v2 DOUBLE;
    -> set v1 = 2*(p1+p2);
    -> set v2 = p1*p2;
    -> insert into result values(v1,v2);
    -> end;
    -> $$
Query OK, 0 rows affected (0.01 sec)

D3_96509_Aditya> delimiter ;
D3_96509_Aditya> call sp_area(10,20);
Query OK, 1 row affected (0.01 sec)

D3_96509_Aditya> select * from result;
+-----------+------+
| perimeter | Area |
+-----------+------+
|        60 |  200 |
+-----------+------+
1 row in set (0.00 sec)

D3_96509_Aditya>
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
D3_96509_Aditya> truncate temp;
Query OK, 0 rows affected (0.02 sec)

D3_96509_Aditya> delimiter $$
D3_96509_Aditya> create procedure sp_num(n1 INT)
    -> begin
    -> declare n2 INT;
    -> declare n3 INT;
    -> set n1 = n1;
    -> set n2 = n1*n1;
    -> set n3 = n2*n1;
    -> insert into temp values(n1,n2,n3);
    -> end;
    -> $$
ERROR 1304 (42000): PROCEDURE sp_num already exists
D3_96509_Aditya>
D3_96509_Aditya> delimiter ;
D3_96509_Aditya> call sp_num(3);
Query OK, 1 row affected (0.00 sec)

D3_96509_Aditya> select * from temp;
+------+------+------+
| n1   | n2   | n3   |
+------+------+------+
|    3 |    9 |   27 |
+------+------+------+
1 row in set (0.00 sec)

D3_96509_Aditya>
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
D3_96509_Aditya> set @num = 44;
Query OK, 0 rows affected (0.00 sec)

D3_96509_Aditya> call fah(@num);
Query OK, 0 rows affected (0.00 sec)

D3_96509_Aditya>
D3_96509_Aditya> select @num;
+-------+
| @num  |
+-------+
| 111.2 |
+-------+
1 row in set (0.00 sec)

D3_96509_Aditya>
```

```
D3_96509_Aditya> delimiter ;
D3_96509_Aditya>
D3_96509_Aditya> set @num = 101;
Query OK, 0 rows affected (0.00 sec)

D3_96509_Aditya> call cel(@num);
Query OK, 0 rows affected (0.00 sec)

D3_96509_Aditya>
D3_96509_Aditya> select @num;
+-------------+
| @num        |
+-------------+
| 38.88888885 |
+-------------+
1 row in set (0.00 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
D3_96509_Aditya> drop procedure inches;
Query OK, 0 rows affected (0.01 sec)

D3_96509_Aditya> drop table if exists show1;
Query OK, 0 rows affected (0.01 sec)

D3_96509_Aditya> create table show1(n1 INT,n2 INT,n3 INT);
Query OK, 0 rows affected (0.01 sec)

D3_96509_Aditya>
D3_96509_Aditya>
D3_96509_Aditya> delimiter $$
D3_96509_Aditya> create procedure inches(n1 INT)
    -> begin
    -> declare n2 INT;
    -> declare n3 INT;
    -> set n2 = n1/36;
    -> set n3 = (n1%36)/12;
    -> set n1 = ((n1%36)%12);
    -> insert into show1 values(n2,n3,n1);
    -> end;
    -> $$
Query OK, 0 rows affected (0.00 sec)

D3_96509_Aditya> delimiter ;
D3_96509_Aditya>
D3_96509_Aditya> call inches(124);
Query OK, 1 row affected (0.00 sec)

D3_96509_Aditya> select * from show1;
+------+------+------+
| n1   | n2   | n3   |
+------+------+------+
|    3 |    1 |    4 |
+------+------+------+
1 row in set (0.00 sec)

D3_96509_Aditya>
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
D3_96509_Aditya> DELIMITER $$
D3_96509_Aditya> CREATE PROCEDURE sp_check_product(n1 INT, n2 INT)
    -> BEGIN
    ->     DECLARE n3 INT;
    ->     DECLARE n4 VARCHAR(20);
    ->     SET n3 = n1 * n2;
    ->
    ->     IF n3 > 100 THEN
    ->         SET n4 = 'GREATER';
    ->     ELSEIF n3 = 100 THEN
    ->         SET n4 = 'EQUAL';
    ->     ELSE
    ->         SET n4 = 'LESS'; -- Added missing SET here
    ->     END IF;
    ->
    ->     INSERT INTO show1 VALUES(n4);
    -> END;
    -> $$
Query OK, 0 rows affected (0.01 sec)

D3_96509_Aditya> DELIMITER ;
D3_96509_Aditya>
D3_96509_Aditya> CALL sp_check_product(12, 12);
Query OK, 1 row affected (0.01 sec)

D3_96509_Aditya> SELECT * FROM show1;
+---------+
| n4      |
+---------+
| GREATER |
+---------+
1 row in set (0.00 sec)

D3_96509_Aditya>
```